

File Management

Session Objectives

By the end of this session, you'll be able to:

- Explain best practices for project file naming and directory structure.
- Describe how version control can be applied through file naming conventions.
- Develop file naming convention and directory structure for a mock list of project files.
- Apply pre-determined file naming convention and directory structure for an example project.

Best Practices in File Naming

Best practices in file naming

There are three main best practices when naming files. File names should be:

- Human-readable
- Machine-readable
- Consistently named

Human-readable

When determining if a filename is human-readable, consider the following questions:

- Can you look at a filename and, without additional information, know what the file is for? What about in a year?
- Are others able to look at your files and know what they are?
- Are you or others able to easily find files that are needed for a project?

Human-readable

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Human-readable

Human-readable file names are short and complete, and they should include at least 3-5 concise conceptual elements. These can include:

- Date of data creation/collection
- Short description
- Activity
- Location where data was collected
- Editor/creator
- Versioning information (see more on this below)
- Any other relevant information

Machine-readable

When determining if a filename is machine-readable, consider the following questions:

- How will a computer sort your filenames? By date, by number, or by first letter?
- If files need to move from one place to another, will they remain interpretable in the same way?
- Can the files be opened by different operating systems or computers?

Machine-readable

- A big part of organizing files has to do with how machines order characters such as letters and numbers.
- Machine-readability across operating systems and applications can be quite complicated, and is beyond the scope of this course.

Machine-readable

Some general guidelines to follow:

- Ordering begins with the first character, and works its way from left to right.
- Numbers are ordered ahead of alphabetical characters.
- Dashes - are ordered ahead of underscores _, and both are ordered ahead of numbers.
- Putting a dash or an underscore at the beginning of a filename can be a way to force it to appear first in a list.
- While some systems will position capital letters ahead of lowercase, it's not recommended to use letter casing as a way to order names.

Machine-readable

Machine-readable filenames often have the following characteristics:

- Only contain letters in the English alphabet, numbers 0-9, dashes, and underscores.
- They do not use spaces or special characters such as: ! @ # \$ % ^ & * () + = { } [] |.
- Naming elements are separated with underscores and/or dashes.
- They use the date format YYYYMMDD or YYYY-MM-DD.

Activity

How would the following file names be ordered?

- Data-analyzed_20260504.csv
- 01_data-analytic_20260504.csv
- -README.txt
- data-analyzed_20260504.docx

Share your answers in the chat!

Activity

Answer

- -README.txt
- 01_data-analytic_20260504.csv
- Data-analyzed_20260504.csv
- data-analyzed_20260504.docx

Consistency, Likeness, and Importance

Consistency, likeness, and importance

- There is no “right” naming convention, and the most important thing is to be consistent.
- When thinking about which elements you want in a naming convention, consider:
 - Elements that important for your project
 - Similarities and differences play into how files are sorted
 - How these play into which element will come first, second, third, etc. in your naming convention

Example

- **Example:** lldr_mpp_20240723.csv
- **Documentation:**
 - Convention: project_location_collection-date.file-type
 - lldr: Leaf litter decomposition rate
 - mpp: Monk Provincial Park

Don't forget to document!

It is good practice to record your naming conventions, and to define all acronyms, abbreviations, codes, and other notations in your project README file.

Managing Directories

Managing directories

- Directories (or folders) are the units that hold files and other directories.
- They are a way of keep your files organized and easy to find.
- Developing a directory structure **before you begin** a project can help with managing all the files that will be generated.

Managing directories

- Directory structures typically have:
 - A root directory (top-level folder)
 - Subdirectories
 - Relevant files
- Directories are denoted by a slash '/' at the end of their name in these diagrams.

```
| -- Project/  
| | _Data-dictionary.txt  
| | _README.txt  
| | -- Data/  
| | -- Funding/  
| | -- Manuscript/  
| | -- Scripts/
```

Managing directories

Before you begin a research project, it's useful to create a plan of your required directories. To start the process, you can think about things such as:

- Directory names
- Directory contents
- Access permissions

Hierarchy depth

There are two distinct ways you can structure your directories:

- A “shallow” directory structure has minimal nesting of sub-directories.
- A “deep” directory structure contains many (potentially very many) sub-directories.

Hierarchy depth

- Choosing which type of structure you want will depend on:
 - How many files the project has
 - The types of files the project has
 - The size/nature of your research team
 - Personal preference

A shallow structure

```
| -- Project/  
| | _Data-dictionary.txt  
| | _README.txt  
| | -- Data/  
| | -- Funding/  
| | -- Manuscript/  
| | -- Scripts/
```

A deep(er) structure

```
| -- Project/  
| | _Data-dictionary.txt  
| | _README.txt  
| | --Analysis  
| | | -- Location-subsets/  
| | | -- Scripts/  
| | | -- Species-subsets/  
| | -- Data/  
| | | -- Processed/  
| | | -- Unprocessed/  
| | -- Funding/  
| | -- Manuscript/
```

File Naming and Version Control

Version control

- Version control is the systematic tracking of the various versions and growth of your files
- File naming can be a way to keep track of versions, which is known as “manual” version control
- Manual version control lends itself well to administrative documents and manuscripts, but in the right context can work for data
- There are also automated version control systems, but these are beyond the scope of the course.

Examples: Manual version control

Administrative documents

- Version number: manuscript_v01.docx, manuscript_v02.docx, ...
- Editor initials: manuscript_v01_NR.docx, manuscript_v01_NR_JA.docx

Data

- “Stage” of data:
 - llldr_mpp_20240723_RAW.csv
 - llldr_mpp_20240723_CLEAN.csv
 - llldr_mpp_20240723_GO-subset.csv

Activity: Creating a File Management System

Activity

Your supervisor has given you another task - she needs help organizing the Social Cognition and Wellbeing Lab's files! With so many students and research assistants in the research group, things have gotten messy and confusing. She's asked you to organize the files with the structure below, under the project name "social_media_wellbeing"

Activity

In breakout rooms, you'll be given a document that lists several files belonging to the project, and will be asked to create a naming convention and directory structure.

Remember: there is no “right” way to organize and name files. Your file hierarchy and naming structure will depend on the project, the files, your research group's standards, your personal preferences, etc!